The Mechanism of Decay from the Bost–Connes System

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Abstract

The structure, thermalization, and motion of matter have been established on the state space of the Bost–Connes system; the change of identity has so far had no mechanism. We define decay as the change of identity of a stable standing wave, $\chi_j \to \chi_k$, and it proceeds in only one way: the amplitude of the initial state merges into the background, the final state re-forms from the background, and an emitted pair carries the record away. Merging and formation are evaluated at the vertices against the background; the evaluation gives the conservation of electric charge and generation charge; the transit is a process whose position is recorded, with a strength universal over all directions; the direction and rate of decay follow, and the Fermi constant and the muon lifetime are given by the numbers of the framework. The exponential decay law, the β spectrum, parity violation, the lifetime hierarchy, and nucleon stability are explained one by one from the same structure.

Contents

1 The Definition of Decay

Matter is a stable standing wave maintained by cycles in a thermal bath[1]. Propagation repeats in units of cycles, and each cycle consists of three steps in sequence[1, 3]:

$$\cdots \xrightarrow{\text{free evolution}} \cdot \xrightarrow{\text{formation exchange}} \cdot \xrightarrow{\text{pattern reconstruction}} \cdots$$

Free evolution accumulates translation phase along the character direction; formation exchange trades one share of amplitude with the vacuum and is the only step of the cycle that communicates with the background; pattern reconstruction rebuilds the standing wave at a new center by stationary phase. The period is $1/v$, and the cycle rate v is shared by all excitations; mass is the phase per cycle multiplied by the cycle rate, $m = v \times \text{(phase per cycle)}$, and exactly one share of phase is registered per cycle[1]. Propagation is carried by the convolution operator K, diagonal in the character basis, and changes of position are carried by the additive displacement T_a[1, 3].

Matter does not always propagate stably. A free neutron becomes a proton with a lifetime measured in minutes, and a muon becomes an electron after 2.197×10^{-6} seconds; no interaction can keep a neutron a neutron. The change nevertheless follows definite laws: the branch of the τ decaying to a muon is always 0.1739, and the same particle decays at the same rate at any time and any place. What, then, is the cause of decay?

- **Why does decay exist?** Stability is maintained by detailed balance, the amplitude flowing out being taken back every cycle; within a self-consistently maintained structure, where does change come from?

- **What is the mechanism of decay?** Through what channel does the change of identity proceed, and by what rules is it constrained; which changes are allowed, which never occur, and what does the change carry away?

- **What determines the direction and rate of decay?** The spontaneous reverse process has never appeared; lifetimes extend from 10^{-25} seconds to beyond 10^{34} years, across more than sixty orders of magnitude. The uniqueness of the direction and the disparity of the hierarchy both require an origin.

1.1 Zero-Element Transit

Decay is a change of identity: a neutron becomes a proton, a muon becomes an electron. The species of a particle is carried by its character direction[1], and a change of identity is the transfer of a standing wave from one direction to another.

Of the three steps of the cycle, free evolution only accumulates phase and trades no amplitude with the background; pattern reconstruction rebuilds the standing wave within its direction by stationary phase; both steps take place inside the standing wave itself, and its form is unchanged in them. The only step that communicates with the background is formation exchange, in which the standing wave lets amplitude flow out and then takes it back[1]. A change of identity can occur only at formation exchange.

Identity is the form of the standing wave, and its change proceeds in two steps: the original species merges with its amplitude into the background, and the new species then forms from the background. Amplitude merged into the background belongs to no direction, so the position at which merging takes place must carry no species. Every character takes the value zero at the zero element, and the only position in the whole space that carries no species is the zero element, written 0. In the structure of positions the zero element is like any lattice site: the motion of a standing wave reaches it and leaves it; in the structure of species it is a vacancy: merged amplitude is not sent by propagation toward any direction, and merging therefore has no reverse. Motion through positions passes through the zero element as through any other site; species terminate at it. The zero element is the unique point at which the structure of positions and the structure of species disagree.

The operator form of the change follows. The derivation uses the section at the unit-group layer, the space $L^2(\mathbb{Z}/p\mathbb{Z})$, with positions labeled by the elements of $\mathbb{Z}/p\mathbb{Z}$ and δ_a the unit vector at position a. Changes of position are carried by addition and species by multiplication; the mathematical form of these two roles is the prescription of the two operations at the zero element: a displacement acts on the zero element as on any lattice site, addition treating 0 as an ordinary site; $0 \cdot n = 0$ for all n, multiplication treating 0 as the absorbing element, with characters extended to the whole space by $\chi(0) = 0$. Addition and multiplication coexist compatibly on all nonzero sites; the zero element is the unique point at which the two operations disagree.

The section carries two families of basic operators: the convolution K, diagonal in the character basis, scales only along directions and preserves identity exactly; the displacement T_a, $(T_a f)(x) = f(x+a)$, has the additive frequency modes $e_\kappa(x) = p^{-1/2} e^{2\pi i \kappa x/p}$ as eigenvectors,

$$(T_a e_\kappa)(x) = e_\kappa(x+a) = e^{2\pi i \kappa a/p}\, e_\kappa(x)\,.$$

The labels of identity are defined on the multiplicative character basis and the labels of motion on the additive frequency basis, and the two bases belong to the two different operations, neither being diagonal in the other. The displacement is not diagonal in the character basis, and under it one direction develops nonzero components along the other directions; but a displacement is a unitary motion of the whole, emits no carrier, and produces no record, and by the record condition[3] it does not constitute a process of identity change.

The operator form and the order of the change are fixed uniquely by the status of the zero element.

Step 1, compatibility with the multiplicative structure. The operator sought labels position; the zero element is a position, so the form is a diagonal operator on positions, a position function $\theta(x)$; commuting with all multiplicative translations requires θ to be constant on multiplicative orbits. The section has only two multiplicative orbits, the unit group and the zero element, so the compatible operators have the form

$$D = \theta_* \not\Vdash_{(\mathbb{Z}/p\mathbb{Z})^*} + \theta_0 \, |\delta_0\rangle\langle\delta_0| \, .$$

Step 2, nontrivial commutation with displacements. The contribution of the identity component merges with the zero-element component; θ_* is absorbed into the diagonal part of K and changes no direction; the nontrivial part is the zero-element component alone, normalized to $\theta_0 = 1$, with the overall strength absorbed into the cycle rate v (§4.3). Hence

$$\boxed{D_F = |\delta_0\rangle\langle\delta_0|}$$

Step 3, flat normalization. The additive Fourier expansion of δ_0 has equal weight on all frequencies, $\langle e_\kappa|\delta_0\rangle = p^{-1/2}$ for every κ, so the transit has no preference among the frequencies of position and makes no selection among directions; selection rests with the evaluation at the two vertices (§2).

Step 4, vanishing at first order and nonvanishing at second order. The commutator of the displacement with the projection is

$$[D_F,\, T_a] = |\delta_0\rangle\langle\delta_a| - |\delta_{-a}\rangle\langle\delta_0| \, ,$$

whose two terms are absorption into the zero element and emission from it, recording exactly the two steps of merging and re-forming; if the two operations commuted, the commutator would vanish identically and no change would exist. A change of identity connects two directions, the initial standing wave along χ_j and the final one along χ_k, and the amplitude of the change is the matrix element of the change operator between them. The first-order matrix element

$$\langle\chi_k|[D_F, T_a]|\chi_j\rangle = \overline{\chi_k(0)}\,\chi_j(a) - \overline{\chi_k(-a)}\,\chi_j(0) = 0 \, ,$$

each term containing one value at the zero element, vanishes term by term since $\chi(0) = 0$: whatever the initial and final directions, a change completed in one step is identically zero. The composition of two steps is not zero. The first action goes through the absorption term, and the amplitude of the initial direction at position a enters the zero element; the second goes through the emission term, and the amplitude of the zero element is sent to position $-b$; the composite matrix element is $-\overline{\chi_k(-b)}\,\chi_j(a)$. A character has modulus one at every site of the unit group, so the modulus of this matrix element is independent of the directions (equal to $1/(p-1)$ for normalized directions): for any pair of directions χ_j, χ_k, the two-step transfer amplitude through the zero element is nonzero.

Zero at first order and nonzero at second order: this is the reason decay exists. Identity cannot be rewritten directly, yet between any two directions there is a two-step transfer amplitude through the zero element. Stability is the norm because of the first-order zero, and decay can occur because of the second-order nonzero; the two coexist in one structure.

A change of identity therefore passes through the zero element at the formation-exchange step and is completed by two vertices, absorption and emission, with one stay at the zero element: first merging into the background, then re-forming. This second-order process is called zero-element transit. Every decay passes through one zero-element transit; the exchange topology of two vertices and one transit is exactly the structure by which the weak interaction proceeds through W exchange[40]. The transit provides the existence of change without selection, the modulus of the composite matrix element being the same for every pair of directions; selection takes place at the two vertices at the ends of the transit.

1.2 The Elements of Decay

A process in the state space is read out through records; the structure of the readout is observational spacetime, in which the process appears as events[3]; the record criterion is applied to transit in §3.1. The picture of the cycle (§1) and the structure of transit (§1.1) together give the definition of decay in four elements.

- **Arena**—the arena of decay is the state space, the same as for formation, thermalization, and motion[1, 2, 3].
- **Object**—the object of decay is a stable standing wave maintained by cycles in the thermal bath, that is, matter itself.
- **Process**—the process of decay is transit through the zero element at the formation-exchange step, acting at most once in each cycle.
- **Record**—decay is completed when the lost amplitude turns into an escaping excitation carrying a record, whose readout is an event in observational spacetime (§3.1).

The complete structure of decay is

$$
\begin{array}{rl}
\text{state space} & \boxed{\begin{array}{c}
\cdots \xrightarrow{\sqrt{1-\varepsilon}\, e^{im/v}} \text{cycle}^{(n+1)} \xrightarrow{\sqrt{1-\varepsilon}\, e^{im/v}} \cdots \\
\wr\ \ \text{transit at formation exchange, share } \varepsilon \\
\chi_j \xrightarrow{\text{absorption vertex}} \text{zero element} \xrightarrow{\text{emission vertex}} \chi_k + \text{emitted pair, record } r
\end{array}} \\
 & \qquad\qquad \wr\ \ \text{readout} \\
\text{observational spacetime} & \boxed{\text{spectral line: position } m\,, \quad \text{width } \Gamma\,, \quad \text{count ratio of branch } r}
\end{array}
$$

The upper box is the whole process in the state space: the cycles advance one by one, each cycle loses the share ε, the loss is distributed through the zero-element transit among the branches r, and the emitted pair escapes with the record; the lower box is the readout in observational spacetime, the position and width of the spectral line and the branch counts. Standard usage divides decays by interaction into strong, electromagnetic, and weak; the decay defined in this paper is the change of identity and corresponds to weak decay, while strong decay, electromagnetic decay, and α decay do not change identity in this framework, and the classification of the three is given together in §6. Which losses are allowed and which forbidden is decided by the evaluation at the transit vertices, the subject of the next section.

2 The Mechanism of Decay

Zero-element transit establishes the existence of change, and the course of a transit is decided by the evaluation at its two vertices; the evaluation gives the rules of transit, and the rules determine the kinds of transit.

2.1 The Rules of Transit

Decay begins with a transit at the formation-exchange step. Of the three steps of the cycle only formation exchange trades amplitude with the background (§1.1); the transit is a second-order process on this step, acting at most once per cycle. One transit replaces one standing wave by a set of standing waves: the amplitude of the initial state merges into the zero element through the absorption vertex, the amplitude during the stay belongs to no direction, and each standing wave of the final state re-forms from the background through the emission vertex (§1.1). The completion of the change is judged by the record: the difference between the initial and final states must be carried away by a really emitted excitation (§3.1), so the final state contains at least two standing waves. Angular momentum is conserved event by event[3]; the standing waves of matter change sign under a 2π rotation and have half-integer spin, one standing wave cannot be replaced by two, and the number of standing waves in the final state is odd; the standing waves emitted in addition to the line therefore come in pairs. The vertex of this paper is the minimal case, exactly one emitted pair; vertices with more emitted pairs are of higher order and lie outside the catalogue.

The standing waves taking part in a transit are identified as particle species by direction; the identification is fixed uniquely in reference [1], and the notation used in this paper is

Direction	Species	Charge	Parity $\chi(-1)$	Remarks
χ_0	background; emergent states condense from it	0	even	condensation direction of baryons
χ_1	up quark	$+2/3$	odd	conjugate pair with χ_5
$\chi_2,\ \chi_4$	charged lepton	-1	even	conjugate pair, the two winding directions of one standing wave
χ_3	neutrino	0	odd	real direction; breaking direction, from which W acquires mass
χ_5	down quark	$-1/3$	odd	conjugate pair with χ_1

In the CRT coordinates $(k \bmod 2,\ k \bmod 3)$ of a direction, the $\mathbb{Z}/2$ component is weak, the $\mathbb{Z}/3$ component is color, and the cross-factor direction is electromagnetic; the lepton conjugate pair $\{\chi_2, \chi_4\}$ carries the two winding directions of one species, while the quark conjugate pair $\{\chi_1, \chi_5\}$ carries two species[1]. Electric charge is carried by direction and is conserved across interactions and transits[1]; a particle and its antiparticle take the same direction with opposite charge, the direction being the identity factor, the distinction between the two being carried by the sign of the charge, and the charge operator being defined on the state space that carries this label.

The transit itself is equivalent for all directions (§1.1); the final state is decided by the vertices. Merging and formation are both evaluated at the vertices against the KMS background[1]; the final states retained by the evaluation constitute the allowed transits, and final states evaluating to zero do not occur. The vertex reads the direction by a character and evaluates it against the background, which is the vertex structure of reference [1]; the complete emission map from the initial state to the line, the emitted pair, and the record is not constructed in this paper, and the rules of this section are the catalogue of that map, taken as the input of the rate evaluation (§4). The rules of transit are given by the symmetry of the background. The symmetry compatible with both propagation and transit is the multiplicative action of the unit group on the lattice sites: the multiplication map μ_c acts as a scalar in the character basis,

$$(\mu_c \chi_k)(x) = \chi_k(cx) = \chi_k(c)\, \chi_k(x)\,,$$

fixes the zero element, $\mu_c \delta_0 = \delta_0$, and commutes with both the convolution operator K and the zero-element projection D_F. The unit group splits by CRT into two factors, $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2 \times \mathbb{Z}/3$, with $\mathbb{Z}/2$ generated by μ_{-1} and $\mathbb{Z}/3$ by μ_2; the $\mathbb{Z}/3$ factor gives the conservation of generation, and the $\mathbb{Z}/2$ factor gives the resolution of the record.

The first conservation law comes from μ_2. The involution ι pairs the lattice sites into doublets $\{a, p-a\}$, the position of a doublet is a generation, and there are three[1]; the three generations of one family are standing waves in the same direction, differing in the doublet on which the lattice projection at the end of formation falls, so that a generation is a record label established at the lattice readout, written j and called the **generation charge**[1]. The doubling map $\mu_2\colon a \mapsto 2a$ links the doublets in turn,

$$\{1,6\} \xrightarrow{\mu_2} \{2,5\} \xrightarrow{\mu_2} \{4,3\} \xrightarrow{\mu_2} \{1,6\},$$

the three generations form a three-cycle, $\mu_2^3 = 1$ ($2^3 = 8 \equiv 1 \bmod 7$), and μ_2 permutes the three lattice configuration states cyclically[1]. The eigensectors of μ_2 are not the generations: characters are eigenfunctions of μ_2, the eigenvalue $\chi_k(2) = \omega^k$ ($\omega = e^{2\pi i/3}$) is merely the $\mathbb{Z}/3$ coordinate of the direction, it is the $\mathbb{Z}/3$ Fourier dual of the lattice position, and position and mode cannot be sharpened at the same time[1]. The generation is carried on the lattice side, the direction does not carry it, and its record is established at the lattice readout (§3.1).

Generation charge is unchanged under propagation; one transit reads out at most one locked link.

Propagation acts on directions: the convolution operator K is diagonal in the character basis, its vertices are scalar actions on directions, carry no lattice operator, and cannot rewrite an established generation record (lattice conservation[1]), $\Delta j = 0$.

μ_2 fixes the zero element, $\mu_2 D_F \mu_2^{-1} = |\mu_2\delta_0\rangle\langle\mu_2\delta_0| = D_F$, $[D_F, \mu_2] = 0$: a move of generation does not pass through the zero element and is not accompanied by emission, so a change of generation can occur only at the lattice projection[1] of the transit; the transit acts at most once per cycle.

The two sides carry the change differently[1]. The inter-generation structure of the quarks contains a locked link, and the lattice projection reads out its share, the Cabibbo parameter λ, the link given in reference [1] lying between the first and second generations; between the generations of the leptons lie free lattice sites with no link to read, and a lepton keeps its generation in a transit. The generation charge flows from the initial state to one standing wave of the final state; that standing wave together with the initial state is called the **line**, and the remaining final standing waves are called the **emitted pair**, which is generation-neutral.

The second structure comes from μ_{-1}. The involution $\iota(a) = p - a$ is multiplication by -1 and maps every direction to itself,

$$(\chi_k \circ \iota)(a) = \chi_k(-a) = \chi_k(-1)\,\chi_k(a)\,,$$

with eigenvalue $\chi_k(-1) = \pm 1$ the parity sign of the direction. -1 is the unique element of order two in the unit group; with a primitive root g, $-1 = g^3$, $\chi_k(g) = e^{2\pi i k/6}$,

$$\chi_k(-1) = e^{2\pi i \cdot 3k/6} = (-1)^k \ :$$

the charged leptons χ_2, χ_4 are even, while the neutrino χ_3 and the two quarks χ_1, χ_5 are odd; conjugate directions have the same sign, $\chi_{-k}(-1) = (-1)^{-k} = (-1)^k$. Write the parity reflection as P, $(Pf)(x) = f(-x)$; then $P\chi_k = \chi_k(-1)\chi_k$, $P\delta_0 = \delta_0$, $PKP = K$, and

$$P\,[D_F, T_a]\,P = [D_F, T_{-a}]\,.$$

Release passes through no zero element and commutes with the reflection; under reflection the transit has its absorption position a and its emission position $-b$ transposed, and the position data of a transit are not invariants of the reflection.

The effect of the reflection falls on the resolution of the record. The zero-element projection is flat over positions (§1.1), and the amplitudes $\chi_j(a)$ of a standing wave of one direction at the lattice sites carry different phases: if absorption were summed coherently over positions, $\sum_a \chi_j(a) = 0$, and no nontrivial direction could be absorbed; if the position of absorption is recorded, the positions add at the probability level, $\sum_a |\chi_j(a)|^2/(p-1) = 1$, the same for every direction. The transit is therefore necessarily a process whose position is recorded, and its strength is independent of the direction: Fermi universality is thereby a corollary of the record (§4.3). The resolution of the record separates two sectors: when the record resolves only the doublet, the two sites of a doublet add coherently, $\chi_j(a) + \chi_j(-a) = \chi_j(a)\,[1 + \chi_j(-1)]$, even directions are absorbed (each doublet contributing 1/3) and odd directions give zero, which is the rule of reference [1] that the entrance must be even, formation recording only the doublet; the transit of the odd directions u, d, ν requires the record to resolve the site within the doublet, the ι-odd position datum. Every decay transit contains an odd direction and therefore carries this datum.

The transit is a process whose position is recorded, with a strength universal over all directions; a transit containing an odd direction records the site within the doublet.

In the readout, the difference of the two sites of a doublet is a component of σ_z type, and ι is a rotation by π about an axis[3], not the mirror; the relation between the mirror and the winding direction is given in §5.3. The operator verification and the process-by-process check are in Appendix A.

The two factors give one item each: the $\mathbb{Z}/3$ factor gives the conservation of generation, and the $\mathbb{Z}/2$ factor gives the resolution of the record; there is no third symmetry compatible with both propagation and transit. The final state of one transit is fixed by the two conservation laws of electric charge and generation charge; the kinds of transit sorted out by the conservation laws are the subject of the next section.

2.2 The Kinds of Transit

Once the two conservation laws are fixed, the final state of one transit and the organization of multiple transits are their corollaries.

Generation charge measures the change. Propagation does not change the generation charge (§2.1), so the change of generation along a path is contributed entirely by the lattice projections of the transits; each transit reads out at most one locked link, and a change spanning k links needs at least k transits. The link given in reference [1] is the λ between the first and second generations: a change of at most one generation is completed in a single transit; the link shares involving the third generation are not among the inputs of this paper (§7.2).

The organization of multiple transits is decided by whether each one is completed. A completed transit is accompanied by emission, and the carrier takes the difference of that transit away and escapes without return (§2.1); once the difference has been carried away, successive transits become distinguishable events, distinguishable degrees of freedom are counted at the probability level[1], and the shares of the transits multiply. An uncompleted transit has no emission and is a virtual process; between virtual processes no carrier tells which came first, they are indistinguishable, indistinguishable degrees of freedom are normalized at the amplitude level[1], and the matrix elements compose in amplitude. These are the only two ways of joining transits, so decay proceeds in exactly three modes: a single transit, a cascade of completed transits, and an amplitude-level composition containing virtual processes.

2.2.1 Single Transit

The final state of one transit is fixed item by item by electric charge and generation charge.

Charge restricts the emitted pair. The emitted pair forms from the background, and its charge is carried by the channel excitation through which the transit passes; in the boson spectrum of reference [1], channel excitations carry only the charges ± 1 (W, the breaking direction χ_3) and 0 (Z, photon, gluon), there is no direction carrying fractional charge, and the charge of an emitted pair is therefore 0 or ± 1. The admissible emitted pairs are thus of two kinds only:

- lepton pairs: $(\ell^-, \bar{\nu}_\ell)$ and (ℓ^+, ν_ℓ) carrying ∓ 1, $(\ell, \bar{\ell})$ and $(\nu, \bar{\nu})$ carrying 0;
- quark pairs: $(u, \bar{d})$ and $(d, \bar{u})$ carrying $\pm(2/3 + 1/3) = \pm 1$, $(q, \bar{q})$ carrying 0.

A mixed pair of one quark and one lepton carries fractional charge and does not occur.

Charge restricts the line. The charge difference between the two ends of the line equals the charge of the emitted pair and must be an integer; the charge differences between a quark and a charged lepton or a neutrino are $\pm 1/3$, $\pm 2/3$, or $\pm 5/3$, all fractional, so there is no transit between quarks and leptons. A change within one species does not change identity. Changes of identity are therefore only of two kinds,

$$\boxed{d \leftrightarrow u\,, \qquad \ell \leftrightarrow \nu_\ell}$$

each with a charge difference of one unit, carried by the emitted pair.

Generation charge fixes the generation. A lepton keeps its generation in a transit (§2.1); in $\ell \to \nu_\ell$ the neutrino is of the same generation as the initial state, the generation charge leaves with the neutrino, and the emitted pair is generation-neutral: the final state of a μ must contain ν_μ, and that of a τ must contain ν_τ. The change $\mu \to e$ between two charged leptons would have to move one unit of generation charge, the lepton side has no lattice projection to carry it, and the amplitude is zero. A quark transit may stay in its generation or shift by one: $d \to u$ stays, $s \to u$ shifts by one generation, the lattice projection carries the share λ, and the probability-level suppression is $\lambda^2 = 0.0505 \approx 1/20$.

All four pairings contain an odd direction (a quark line, a neutrino, or a quark pair), and every transit records the site within the doublet (§2.1).

The kinds of single transit with exactly one emitted pair are thereby exhausted, exactly four:

- quark line with lepton pair, $d \to u + (\ell^-, \bar{\nu}_\ell)$: β decay;
- quark line with quark pair, $d \to u + (d, \bar{u})$: nonleptonic decay;
- lepton line with lepton pair, $\ell \to \nu_\ell + (\ell'^-, \bar{\nu}_{\ell'})$: the decay of the μ and the leptonic branches of the τ;
- lepton line with quark pair, $\ell \to \nu_\ell + (d, \bar{u})$: the hadronic branches of the τ.

The reverse of each ($u \to d$, $\nu_\ell \to \ell$) is equally allowed at the vertex level; whether it occurs is decided by the completion condition of the final state (§2.3). The emitted pairs of the τ are $(e, \bar{\nu}_e)$, $(\mu, \bar{\nu}_\mu)$, and $(d, \bar{u})$ carrying three colors, with branch counts $1 : 1 : 3$. The conservation laws of the four pairings are checked process by process in Appendix A.

W and Z occupy different positions in this structure. W belongs to the breaking direction χ_3 and is odd; each appearance of it is the two-vertex structure of a transit: the decay of W is the emission of a transit, obeys the conservation of charge and generation charge, and its final state is necessarily one of the two kinds of emitted pair. Z belongs to an even direction and is a channel excitation; the excitation returns its weight to the channel without changing the identity of any standing wave, which is called **release**, and passes through no transit; both the visible and the invisible final states are open, and the structure of the width is given in reference [1].

2.2.2 Multiple Transits

Multiple transits divide into two kinds by the level at which they are joined.

The joining of completed transits is a cascade. Each transit is accompanied by its own emission (§2.1), the carrier takes the difference of that transit away, successive transits are mutually distinguishable events, and the probability of reaching a specified final state along the chain is the product of the probabilities,

$$\mathcal{P}(\text{chain}) = \prod_i \mathcal{P}_i \,.$$

A cascade introduces no new kind; each transit in the chain is one of the four. A change across several links proceeds by cascade: each transit reads out one link, the link between the first and second generations carries λ, with probability-level suppression λ^2; leptons cannot move generation (§2.1), so cross-generation cascades are carried entirely by quarks. The final state of a heavier state after one decay may decay again, stepping down until stable matter is reached; cascades of this kind exist on both the quark and the lepton side. A recorded cascade cannot return; the chain only advances.

The joining that contains virtual processes is amplitude-level composition. Between the transits there is no emission and no record, the transits are indistinguishable, and the total amplitude is the product of the matrix elements,

$$A(\text{composition}) = \prod_i A_i \,.$$

The mixing of K^0 and $\bar{K}^0$ is this process[36]: the two lines $d \to s$ and $\bar{s} \to \bar{d}$ each span one generation and each pass through one transit, both transits are virtual, the four vertices compose into one amplitude-level process, the mixing amplitude is of the order of the square of a single amplitude, and the smallness of the mixing follows. An unrecorded composition is reversible, so K^0 and $\bar{K}^0$ oscillate back and forth, in contrast to the irreversibility of a cascade.

The kinds of transit of the minimal vertex are thereby exhausted. Changes of identity are only $d \leftrightarrow u$ and $\ell \leftrightarrow \nu_\ell$, emitted pairs are only lepton pairs and quark pairs, cross-generation changes proceed by cascade, and unrecorded compositions appear only at the amplitude level; the decay of emergent states is the subject of the next section.

2.3 Decay Branches

The kinds of transit give the changes allowed at the vertex level; for a change to become a decay, the final state must also be completable: every standing wave of the final state has to complete its own formation cycle, so the sum of the final masses is smaller than the initial mass. For a free initial state, the reverses of the four transits ($u \to d$, $\nu_\ell \to \ell$) have heavier final states and are closed by this; the reverses inside a composite body are assessed for the specific initial and final states ($\pi^\pm$, §4.7). The allowed transits together with the completable final states give all the decay branches and the stable matter.

The condensate body of an emergent state is maintained by the pole, does not cycle, and the reading of complex energy (§3.3) does not apply to it directly[1]. A baryon is not a bound state of three quarks but a single collective condensation in the χ_0 direction; the readings of its constituents are internal projections along the nontrivial character directions, and the shares of χ_1 and χ_5 are the up- and down-quark constituents[1]. The decay of an emergent state is therefore a transit of an internal projection, and the conservation laws and the completion condition of the preceding two sections apply item by item. The internal projection of the neutron has exactly one open branch, $d \to u$ (§2.2); the neutron–proton mass difference $\Delta m = 1.30\,\mathrm{MeV}$ exceeds m_e[1], and phase space remains after the electron is completed; for the proton the same branch runs in reverse, the final state cannot be completed, and the branches in which a quark becomes a lepton or a neutrino are closed by charge (§2.2). Emergent and elementary states are unified in the mechanism of decay; the magnitude of the neutron decay is given in §5.5.

The two conservation laws, the three modes, and the completion condition give the complete table of branches at $p = 7$:

Initial state	Final state	Identity change	Mode
n	$p\,e^-\bar\nu_e$	$d \to u$, same generation	single; unique branch
μ^-	$e^-\bar\nu_e\,\nu_\mu$	$\mu \to \nu_\mu$	single; unique branch
τ^-	$\nu_\tau + \{e\bar\nu_e,\ \mu\bar\nu_\mu,\ d\bar u \times 3\}$	$\tau \to \nu_\tau$	single; three kinds, counts 1 : 1 : 3
$s \to u$ type	strange decays	$s \to u$, shift of one generation	single; share $\lambda^2 \approx 1/20$
heavy-state chains	stepwise downward	multiple	cascade; probabilities multiply
$K^0 \leftrightarrow \bar K^0$	mixing oscillation	$d \to s$ and $\bar s \to \bar d$	amplitude-level composition; four vertices, reversible
$\pi^\pm$	$\ell\,\bar\nu_\ell$	$u \to d$, annihilating with $\bar d$	single; winding seed, e channel suppressed (§4.7)
W	$\ell\bar\nu_\ell \times 3$, $q\bar q\,' \times 6$	emission of a transit	single; nine branches, counts 3 : 6
Z	visible and invisible	no identity change	resonance release; bypasses the zero element

Every row of the table is sorted out uniquely by the conservation laws (§2.1), the kinds (§2.2), and the completion condition.

States with all branches closed constitute stable matter.

- The proton. Its condensate body is the lowest color singlet, the condensate layer has no lower final state, and the constituent directions have no χ_0 matrix element[1]; among the branches of the internal projections, those in which a quark becomes a quark have heavier final states, and those in which a quark becomes a lepton or a neutrino have fractional charge differences; the two kinds are closed separately.
- The electron. The lightest charged stable state; under charge conservation there is no charged final state of lower mass.
- The neutrino. The lightest fermion, electrically neutral, with no insertion flow[1].
- The photon. Massless, carrying no cycle, with no share to lose.

Each forbidden process has a definite source of closure: $\mu \to e\gamma$ is closed by the conservation of generation charge, $\mu \to 3e$ by the absence of links between lepton generations, emitted pairs of type $(q, \bar\ell)$ by charge conservation, all branches of the proton by the completion condition and charge, and n–$\bar n$ oscillation by the conservation of the net number of lines at the vertices (each vertex continues one line and emits one pair, and $\Delta B = 2$ has no carrier)[1]. The allowed changes and the closed states come from the same set of vertex identities. By what laws the branches proceed is the subject of the next section.

3 The Records of Decay

A transit replaces one standing wave by a set of standing waves (§2), and the process takes place in the state space; for it to become an event in observational spacetime it must be recorded (§1.2). The record criterion applied to transit gives the condition for a transit to become an event, the counting of events, and the readings of an event.

3.1 The Record Criterion

A record is given by the conjunction of three conditions[3]: the recorded difference is carried by classical quantum numbers, spectral classes commuting with propagation or ι-symmetric lattice quantities that are targets of φ-preserving conditional expectations[1]; the difference is carried away by a really emitted excitation; the carrier escapes without return. Each of the three takes a definite form on the transit.

The first condition, the carrying quantities. Propagation is carried by the convolution operator K, diagonal in the character basis, $K\chi_k = L_k\chi_k$ with $L_k = L(\beta, \chi_k)$[1]; write $\Pi_k = |\chi_k\rangle\langle\chi_k|$ for the projection onto the direction χ_k. The difference between initial and final states is carried by three quantities. Electric charge takes its value by direction (the table of §2.1),

$$Q = \sum_k q_k \, \Pi_k \, , \qquad QK = \sum_k q_k L_k \, \Pi_k = KQ \, ,$$

$[Q, K] = 0$. Generation charge is the doublet-position label at the lattice readout, and its carrying quantity is the ι-even lattice projection,

$$E_j = |\delta_{a_j}\rangle\langle\delta_{a_j}| + |\delta_{-a_j}\rangle\langle\delta_{-a_j}| \, , \qquad G = \sum_j j \, E_j \, , \qquad j \in \mathbb{Z}/3 \, ,$$

with $\{a_j, -a_j\}$ the jth doublet; the sum over the two sites of a doublet is a target of a φ-preserving conditional expectation and a legitimate readout[1]. Propagation acts on directions and not on the readout label (lattice conservation[1]), and G is unchanged in propagation. The parity reflection P is diagonal in the character basis, $P\chi_k = (-1)^k\chi_k$ (§2.1), $P = \sum_k (-1)^k \Pi_k$, $[P, K] = 0$.

The three are unchanged in propagation and take definite values at the lattice readout, charge in $\{0, \pm 1/3, \pm 2/3, \pm 1\}$, generation charge in $\mathbb{Z}/3$, parity in ± 1; the effective overlap of a record takes only the values 0 and 1[1, 3], and all three are classical quantum numbers. The winding direction is not among them: the readout takes Galois-conjugation-invariant combinations[1], the two winding directions of a conjugate pair are indistinguishable in the readout, the difference of winding direction does not enter the record, and the identity record of a lepton therefore contains only even components (§2.1). The differences of the three quantum numbers across the transit are taken up by the final state: the charge difference is carried by the emitted pair, the generation charge continues along the line or moves across a link, the parity sign of each direction is recorded with the direction, and the lattice position of the transit enters the record (§2.1).

The second condition, emission. The difference must be carried away by a really emitted excitation. Each standing wave of the final state forms from the background at the emission vertex; formation is the repeated application of propagation and is completed when the standing wave has built its own cycle[1]; the mass of each final standing wave is read from its cycle, and the condition for formation to be completed is that the energy of the initial state suffices to complete all final states,

$$\sum_{\text{out}} m < m_{\text{in}} \, ,$$

the completion condition of §2.3. A completed standing wave propagates by its own cycle[3] and is a real emission; the record of an elastic process is carried by a photon[2], and the record of a decay is carried by the emitted standing waves themselves. When the final state cannot be completed, emission fails, the transit is not completed, and it enters only the amplitude (§2.2). The reality of the emission and the completion of the final state are one and the same condition.

The third condition, irreversibility. The carrier escapes without return. The completed final standing waves leave the emission vertex along their own cycles, and their overlap with the source decays to zero under propagation[3]; a return would require the emitted pair to become coherent again with the source at the same vertex, the overlap is zero, and the record cannot be undone. The structural origin of the absence of return is the zero element: amplitude merged into the zero element is not sent by propagation toward any direction, and merging has no reverse (§1.1), developed in §4.

The conjunction of the three conditions is the completion of the transit:

The record of a decay consists of the differences of charge and generation charge carried by the final standing waves and of the lattice position of the transit; with the final state completed and the carrier escaped the transit is complete, and lacking either it is a virtual process.

A completed transit is read out in observational spacetime as one event; the counting and the readings of events are the subject of the next two subsections.

3.2 Records and Events

A completed transit is read out in observational spacetime as one event; an uncompleted transit is a virtual process and has no event. One event carries three items:

- the recorded difference, that is, the branch r: the kind of the emitted pair and the shift of generation charge;
- the cycle number n at which it occurred;
- the energies and momenta of the final standing waves, carried by the phase space of the final state (§4).

Events are counted in units of cycles. The transit acts at most once per cycle (§2.1); the probability of completion is written ε and called the **loss share**; ε is fixed by the vertex evaluation, is constant from cycle to cycle, and its value is given in §4. Survival has no record and proceeds at the amplitude level: the survival amplitude of one cycle is the cycle eigenvalue

$$\Lambda = \sqrt{1-\varepsilon}\, e^{im/v}$$

(the structure diagram of §1.2), the survival amplitude over n cycles is Λ^n, and the survival probability is $|\Lambda|^{2n} = (1-\varepsilon)^n$. The probability that the event occurs in the nth cycle is

$$\mathcal{P}(n) = (1-\varepsilon)^{n-1}\,\varepsilon\,, \qquad \sum_{n\geq 1} \mathcal{P}(n) = 1\,, \qquad \bar{n} = \frac{1}{\varepsilon}\,,$$

a standing wave with $\varepsilon > 0$ eventually decays, with mean lifetime $1/\varepsilon$ cycles; the distribution depends only on ε and not on the number of cycles already survived, and the event has no memory (§5.1).

Events of different branches carry different records, are mutually distinguishable events, and add at the probability level,

$$\varepsilon = \sum_r \varepsilon_r \, ,$$

with r running over the branches of the initial state in the table of branches (§2.3). Virtual processes do not count toward ε: an uncompleted transit has no record, its amplitude enters the surviving standing wave and belongs to the correction layer[1], accounted for in §4.4. The two levels of the readout are thereby fixed: events are counted at the probability level and survival proceeds at the amplitude level[1]; the three readings of an event are the subject of the next subsection.

3.3 The Readings of an Event

An event has three kinds of reading, the width of the spectral line, the position of the spectral line, and the count ratio of the branches (the structure diagram of §1.2), each derived from the cycle.

The width comes from the counting of events. Convert the survival probability $(1-\varepsilon)^n$ to time: one lattice step per cycle, of length $1/v$, the same for all standing waves[3], so $n = vt$. Survival probabilities multiply over successive intervals, $S(t_1+t_2) = S(t_1)\,S(t_2)$, and the monotone solution is the exponential,

$$N(t) = N_0\,(1-\varepsilon)^{vt} = N_0\,e^{-\Gamma t}\,, \qquad \Gamma = -v\log(1-\varepsilon) = v\,\varepsilon + O(\varepsilon^2)\,,$$

so the exponential decay law[5] is a theorem. The shape of the line is given by the resolvent of the sum over cycles, the Fourier sum over the cycle count at energy E,

$$A(E) = \sum_{n\geq 0} \Lambda^n e^{-iEn/v} = \frac{1}{1-\Lambda\,e^{-iE/v}}\,,$$

convergent since $|\Lambda| < 1$. Near $E \approx m$, $\Lambda\,e^{-iE/v} = \sqrt{1-\varepsilon}\,e^{i(m-E)/v} \approx 1 - \varepsilon/2 + i(m-E)/v$,

$$A(E) \approx \frac{v}{\Gamma/2 + i(E-m)}\,, \qquad |A(E)|^2 \;\propto\; \frac{1}{(E-m)^2 + \Gamma^2/4}\,,$$

the Breit–Wigner distribution[9]; the line center reads the mass, and the line width reads the lifetime.

The mass comes from the surviving amplitude. Survival is at the amplitude level; the modulus of the eigenvalue Λ is given by ε and its phase by the phase per cycle m/v (§1, the frequency-readout lemma[1]); taking the logarithm and multiplying by the cycle rate,

$$v\log\Lambda = i\,m - \frac{\Gamma}{2}\,,$$

the imaginary part is the mass and the real part is half the width; the one half in front of Γ comes from the difference between the two levels, loss being counted at the probability level while the surviving amplitude bears only half of the logarithm. The two readings come from one eigenvalue, the complex energy

$$\omega = -iv\log\Lambda = m + \frac{i\Gamma}{2}$$

being the pole of the resolvent[31]; with energies written as e^{-iEt} one takes the conjugate, $E = m - i\Gamma/2$, the usual form of the Breit–Wigner distribution. The domain condition of the mass readout, $m < \pi v$[1], and the condition of the width readout, $\varepsilon \ll 1$, are independent of each other; the modulus has no lattice multivaluedness, and the width adds no new domain condition.

The count ratio of the branches comes from the distribution of the loss. Branches are mutually distinguishable events (§3.2),

$$\mathrm{BR}_r = \frac{\varepsilon_r}{\varepsilon} ,$$

a branching ratio is a ratio of shares and a structural quantity. The share of each branch factorizes into three parts: the evaluation of the two vertices (strength absorbed into v), the count of the branch, and the phase space of the final state; the phase space is a power of (m/v), the exponent equal to the number of free final-state energies, and its continuous form is given by the inheritance of the spacetime structure (§4.2).

The three readings are now complete: the width from counting, the mass from survival, and the branching ratio from distribution, all given by the cycle eigenvalue and the distribution of the loss. Why decay runs in one direction only is the subject of the next section.

4 The Rate of Decay

The rules, kinds, and records of transit are established; what remains is the third question of the first section, what determines the direction and rate of decay. The loss runs in one direction only, and the loss share per cycle is converted into a continuous rate; the evaluation of rates proceeds in four layers, the formula, the calibration by the muon, the dwell correction for branches containing quarks, and the application case by case.

4.1 The Irreversibility of Decay

The direction of time has been established separately at the three levels of structure, ensemble, and event[1, 2, 3]; the one-way character of decay belongs to the rate of a single process. The dynamics itself is time-reversal symmetric: σ_t is unitary and invertible, and the transit matrix element satisfies under conjugation

$$[D_F, T_a]^\dagger = -[D_F, T_{-a}] ,$$

so the forward and reverse vertices have equal modulus, and the asymmetry does not come from the vertices.

The asymmetry appears in the rates. On the forward side, the loss share is nonzero factor by factor,

$$\varepsilon = 2\pi \sum_r |a_r|^2 \, \tilde{\rho}_r ,$$

with a_r the vertex amplitude of branch r and $\tilde{\rho}_r$ the density of final states (§4.2): admissible branches exist, and their vertex amplitudes are given by the evaluation at the two ends (§2); the density of states is positive when the final state can be completed (§2.3). Neither factor vanishes, and $\Gamma > 0$. On the reverse side, the rate of a spontaneous reverse process is proportional to the occupation on the incoming side; after the loss the local occupation is zero, and by three facts it does not rise again: thermal activation vanishes at zero temperature, vacuum freezing[2]; the carrier escapes without return, a controlled reversal would require gathering the escaped carriers, and this is excluded by the replacement of the environment[2] (§3.1); the outgoing modes are moved into the complement of the environment, the dilation is a bilateral shift with continuous spectrum, and the occupation does not recur[2]. The escape of the carrier cannot be undone, and the reverse rate is identically zero.

The one-way character is borne algebraically by the zero element. Addition is a group, every displacement has an inverse, $T_a T_{-a} = \mathbb{1}$, and a change of position can be retraced; multiplication is a semigroup, the zero element absorbs everything, $0 \cdot n = 0$ for all n, and no multiplication sends the zero element back to a nonzero element. The zero element is the unique point at which the two operations disagree, and the existence of decay comes from it (§1.1); the one-way character of decay comes from the one-sidedness of the absorbing element: all other links of the process are reversible, propagation, displacement, and the two mutually conjugate vertices, and the only one-sided link is the merging into the zero element: merged amplitude is not sent by propagation toward any direction and has no inverse under multiplication, while the re-formation from the zero element proceeds through the emission term of the displacement and is not the reverse of merging. The zero element has no inverse under multiplication, and the loss in decay has no return flow: both come from one semigroup property.

Irreversibility does not belong to the dynamical equation but to the boundary between the process and its environment: the equation is reversible, the boundary one-way. The two facts involving the environment come from the replacement of the environment in reference [2], and the third is the spectral structure of the dilation; the reverse vertices exist with equal modulus, and only the spontaneous reverse process is excluded. The relation to thermalization is fixed thereby: thermalization relaxes through the zero element at the ensemble level, and the change of identity is the slowest tier of the relaxation catalogue[2]; decay changes identity through the same zero element at the level of a single system, proceeds even at zero temperature, is a coherent process of the dilation layer, and its rate is given by the modulus deficit of the eigenvalue with no temperature factor. The direction is established; the numerical values of the rates are the subject of the following subsections.

4.2 The Rate Formula

The cycle map with transit is

$$M = \mathcal{U}\big(\nVdash - \mathcal{E}_{\text{form}} - \mathcal{E}_{\text{transit}}\big) ,$$

the first two terms maintaining the standing wave (§1) and the third being the loss; a decaying state is an eigenmode of M. The sum over cycles is the geometric series of the resolvent,

$$\sum_{n \geq 0} \Lambda^n = \frac{1}{1 - \Lambda} ,$$

the same mechanism as the sum over self-loops in the mass-evaluation rules of reference [1]; the sum of the series has its pole at the complex energy of §3.3 ($E = m - i\Gamma/2$ in the e^{-iEt} convention), and energy conservation is an output of the summation. The modulus deficit per cycle is given by the vertex amplitudes and the density of final states in the weak-coupling limit,

$$\varepsilon = 2\pi \sum_r |a_r|^2 \, \tilde{\rho}_r ,$$

the golden rule in discrete time, which this paper takes as a matching condition: a_r is the vertex amplitude of branch r, dimensionless; $\tilde{\rho}_r$ is the density of states per unit of the dimensionless energy E/v, $\tilde{\rho}_r = v\rho_r$. The strength of the transit is absorbed into v (§4.3), the contact matrix element is $M_r = v\, a_r$, and $\Gamma = v\varepsilon = 2\pi \sum_r |M_r|^2 \rho_r$ is the golden rule in continuous time. On returning to relativistic normalization, the normalization $\sqrt{2E}$ of each final line, the flux factor $1/(2m)$, and the Lorentz-invariant phase space are all inherited from the spacetime structure[1]:

$$\boxed{\Gamma = v\,\varepsilon = \frac{1}{2m} \int |M|^2 \, d\text{LIPS}}$$

The time integral appears only once, in the sum over cycles, with no double counting; for small shares $\Gamma = v\varepsilon$ agrees with the exact reading $-v\log(1-\varepsilon)$ (§3.3). ε is the modulus deficit of the cycle eigenvalue per cycle, that is, the probability that a record occurs in that cycle; the matching condition holds for weak coupling, a continuous spectrum of final states, and cycle numbers large compared with one and small compared with the recurrence time. The formulas below are the integrals, over the inherited phase space, of the contact matrix element of the framework (the $V-A$ form, §5.3; the spin structure from reference [3]); the results of the integration agree with the standard forms[12].

4.3 The Muon Lifetime

The strength of the transit is absorbed into the cycle rate v (the normalization of §1.1) and is the same for all directions (§2.1); the dimensionless coefficient of the contact matrix element is matched to the experimental definition $v = (\sqrt{2}\,G_F)^{-1/2}$, which gives

$$G_F = \frac{1}{\sqrt{2}\,v^2} = 1.16665 \times 10^{-5}\,\mathrm{GeV}^{-2}\,,$$

deviating from the experimental value[41] by $+0.023\%$. The unique branch of the muon gives

$$\Gamma_\mu = \frac{G_F^2 m_\mu^5}{192\pi^3}\, f\!\left(\frac{m_e^2}{m_\mu^2}\right)\,, \qquad f(x) = 1 - 8x + 8x^3 - x^4 - 12x^2\log x\,,$$

with f the phase-space integral of the three-body final state and the masses taken as the spectral values computed in reference [1]. Numerically

$$\tau_\mu = 2.18709 \times 10^{-6}\,\mathrm{s}\,,$$

a deviation of -0.45%[41]. The residual is attributed to the electromagnetic correction of the charged lepton lines: the dwell of the two charged lines, μ and e, through the electromagnetic channel is not derived in this paper; in the standard calculation this share is $\frac{\alpha}{2\pi}\left(\frac{25}{4} - \pi^2\right) = -0.42\%$[15], comparable to the residual. The calibration is complete; branches containing quarks require in addition the dwell correction, the subject of the next subsection.

4.4 The Dwell Correction

The residence of a lepton (its character direction[1]) forms in one step from a single generator, and the record is established as soon as it is emitted; the dwell of a charged lepton line with the electromagnetic channel passes through no broken segment, and its form is not derived in this paper (§4.3). The formation of a quark pair passes through the color channel and has a stretch of dwell, its endpoints not closed and its labels not fixed, and the correction lives on this broken segment (the stretch of the formation path on which the record is not yet established[1]).

Step 1, exactly one composite per segment. The correction of a broken segment is a two-vertex composite across the segment; the transition side of the segment is the passage through the zero element, instantaneous and without dwell, and a repeated composite has no place on the same segment: each broken segment carries exactly one composite[1].

Step 2, the catalogue of insertion channels. A channel taking part in the composite must have a massless carrier, that is, color or electromagnetism. The weak channel is broken, the carrier mass confines it within the subnuclear range [1, 4], and it provides no massless insertion; the change of identity itself is carried by the zero-element transit (§1.1). The catalogue has two entries only.

Step 3, configurations and shares. The configuration space of the composite consists of the combinations of the insertion-pair member and the line, and the four configurations have equal weight: the two members of the insertion pair are Galois conjugates, $|R_2| = |R_4|$, the two lines have the same direction, and the normalization is shared equally over the unitary displacement algebra, independent of the channel charge. The vertex coupling is proportional to the channel charge, and a color-neutral and electrically neutral intermediate residence (the direction χ_3) makes the contribution of half of the configurations zero, the generalization of the electric-neutrality exemption[1]: the share of nonzero configurations for each kind of composite is $\frac{1}{2}$. The configurations are counted one by one in Appendix B.

Step 4, the counting of labels. The generators of a color composite carry color labels, and the intermediate state sums over the $d_2 = 3$ color labels released; the generators of an electromagnetic composite carry no color label, and the intermediate state is a single term. The total share of a single composite is

$$\xi = \tfrac{1}{2} d_2\, \alpha_s^{(0)} + \tfrac{1}{2}\, \alpha_{\mathrm{EM}}^{(0)} = 0.176343\,,$$

where $\alpha_s^{(0)} = |R_1|^2/3 = 0.11492$ and $\alpha_{\mathrm{EM}}^{(0)} = |R_2|^2/6 = 0.0079270$ are the Born-level couplings: couplings of real processes take corrected-level observed values and inputs inside the correction layer take Born-level values (the evaluation rule of reference [1]); they are built from the resonance ratios R_k computed in reference [1].

Step 5, nesting. The intermediate state of the composite is itself a broken segment: the intermediate pair propagates in dwell through the color channel between the two vertices, its endpoints not closed and its labels not locked, agreeing item by item with the definition of a broken segment. By Step 1 it carries exactly one composite of its own; the recursion terminates only at a record. The broken segment is a decoherent segment[1], the shares of the layers multiply at the probability level and are summed, by the same mechanism as the dwell series of the self-loops in reference [1], entering the weight of the width linearly, and the sum has the same form as the resolvent of the sum over cycles (§4.2):

$$1 + \xi + \xi^2 + \cdots = \frac{1}{1-\xi}\,, \qquad \boxed{w_{\mathrm{had}} = \frac{d_2}{1-\xi} = 3.642}$$

The form of the correction is a nested resolvent, layer upon layer; the catalogue of insertion channels is the same at every layer, because the object of the recursion is the segment, and the properties of a segment do not depend on the kind of composite occupying it.

4.5 τ Decay

The weights of the three kinds of τ branch are $f_e : f_\mu : w_{\mathrm{had}}$ (the hadronic channel being inclusive, containing the $s\bar{u}$ share with $|V_{ud}|^2 + |V_{us}|^2 = 1$, so that the count 3 is unchanged), where $f_\ell = f(m_\ell^2/m_\tau^2)$ is the phase-space factor, $f_e = 0.9999993$, $f_\mu = 0.972564$; the correction of the hadronic branch is given by the dwell (§4.4), and the total width is

$$\Gamma_\tau^{\mathrm{tot}} = \frac{G_F^2 m_\tau^5}{192\pi^3}\left[f_e + f_\mu + w_{\mathrm{had}}\right],$$

with m_τ the spectral value computed in reference [1]. The numbers compared with experiment:

Quantity	This paper	Experiment	Deviation
$\Gamma(\tau \to e\bar{\nu}_e\nu_\tau)$ [GeV]	4.050×10^{-13}	4.040×10^{-13}	$+0.25\%$
$\Gamma(\tau \to \mu\bar{\nu}_\mu\nu_\tau)$ [GeV]	3.939×10^{-13}	3.943×10^{-13}	-0.10%
$\mathrm{BR}(\tau \to \mu\bar{\nu}_\mu\nu_\tau)$	0.1732	0.1739	-0.40%
τ_τ	2.894×10^{-13} s	2.903×10^{-13} s	-0.30%

The two leptonic partial widths, like τ_μ, do not contain the electromagnetic correction of the charged lepton lines (§4.3); the nesting corrects the hadronic side only, and the residuals of the leptonic branches are retained as they are in the branching ratio and the total width.

4.6 β Decay

The vector and axial-vector readouts have different standing. The vector readout runs along the continuation of the line, and the commutator of μ_2 (§2.1) protects it: insertions are diagonal in the character basis and commute with μ_2, the vector weight is not renormalized by the strong dwell and is identically one, which is the conservation of the vector current[12], consistent with the constancy of the Ft values of superallowed $0^+ \to 0^+$ transitions[44]. The share remaining in the branch of the same generation is the lattice projection

$$|V_{ud}| = \sqrt{1-\lambda^2} = 0.97443\,,$$

a deviation of $+0.078\%$, with λ the value computed in reference [1].

The axial-vector readout has no commutator protection: its pairing is selected by the winding structure of the vertex (§5.3) and does not run along the charge flow, so the axial component is corrected by the dwell.

Step 1, cancellation in the ratio. The axial-vector weight g_A is the ratio of the axial-vector and vector readouts; on the vector side each insertion evaluates to the identity (commutator protection), and any insertion acting equally on the two readouts cancels in the ratio. The electromagnetic composite carries no lattice operator and changes no species, acting equally on both readouts: only the color composite remains in the ratio.

Step 2, the series. The axial insertions are self-energy composites on the line, both vertices falling on the dwell segment and occupying no position on the transition side, so the one-composite rule of Step 1 of §4.4 does not apply to them; the line of a nucleon is maintained by the emergent state, the dwell does not terminate, the insertions of successive layers accumulate in phase, and the series has the same form as the nesting (§4.4):

$$\sum_{n\geq 0}\left(\tfrac{1}{2}d_2\,\alpha_s^{(0)}\right)^n = \frac{1}{1-\tfrac{1}{2}d_2\,\alpha_s^{(0)}} = 1.2083\,.$$

On the vector side each insertion evaluates to the identity, the weight is not changed by the series and remains one: the conservation of the vector current survives the series.

Step 3, vertex contact. The product of the line of a conjugate pair with an insertion member can fall on an odd direction: $\chi_1\bar{\chi}_4 = \chi_3$, the landing point being the direction of the vertex itself; the configuration belongs to the vertex, is an amplitude-level projection within the vertex[1], each readout has exactly one share of it, and it does not enter the series. The intermediate state is colorless, color conservation restricts the configuration to the color-singlet component, and the amplitude projection is $1/\sqrt{d_2}$; the coupling is taken at Born level (the evaluation rule[1]). A lepton line has no such configuration ($\chi_2\bar{\chi}_4 = \chi_4$, not on an odd direction): the axial-vector weight of a lepton is one, in agreement with observation.

Step 4, assembly.

$$\boxed{g_A = \frac{1}{1-\frac{1}{2}d_2\,\alpha_s^{(0)}} + \frac{\alpha_s^{(0)}}{\sqrt{d_2}} = 1.2083 + 0.0663 = 1.2746}$$

The experimental value is 1.2754, a deviation of -0.06%.

4.7 π and W

The decay of $\pi^{\pm}$ is controlled by the winding seed: the vertex at which an initial state of spin zero emits a $V{-}A$ pair (§5.3) vanishes identically in the massless limit, and the first nonzero term is seeded by the cycle phase of the final lepton,

$$\Gamma \;\propto\; m_\ell^2\left(1-\frac{m_\ell^2}{m_\pi^2}\right)^2 ,$$

so helicity suppression is a corollary of the $V{-}A$ contact form (§4.2) for a two-body final state. The structure of the branching ratio $R_{e/\mu}$[42] is reproduced thereby; its numerical residual is amplified from the -0.9% deviation of the framework value of m_π, and the improvement lies on the side of reference [1].

The nine branches of the W give the structural value $\mathrm{BR}(W\to\ell\nu_\ell) = 1/9 = 0.111$ (experiment 0.109). The dwell correction of the hadronic branches cannot take over the values at the τ scale: in the framework the couplings do not run with the energy scale, and what changes with the scale is the equal-share counting of the composite shares[1]. That counting law is the remaining unfinished part of this paper; its derivation requires order-by-order counting at the amplitude level and lies beyond the scope of this paper; the evaluations of §4.5 and §4.6 do not depend on it.

5 The Phenomena of Decay

§1.1 to §4 have established transit, rules, records, and rates. The phenomena of decay run from the unpredictability of the instant of a single nuclear decay to the stability of atomic nuclei; five phenomena, from the simplest to the most complex, each test one link of the mechanism, and the stability of nuclei consists of the closure of transits.

5.1 The Exponential Decay Law

The instant at which a single nucleus decays cannot be predicted: it may decay in the next second or after a thousand years; and a nucleus that has already survived a thousand years has the same distribution of remaining lifetime as a freshly prepared one. On the ensemble side, the count falls off exponentially[5], and the rate is the only definite quantity. Memorylessness and the absence of an instant are derived from the structure below.

Memorylessness is the constancy of the loss share. The loss share is the same ε in every cycle (§3.2), and the conditional probability of surviving k further cycles after surviving n is

$$\frac{(1-\varepsilon)^{n+k}}{(1-\varepsilon)^n} = (1-\varepsilon)^k ,$$

independent of n: survival does not depend on the time already survived, and old samples obey the same distribution as freshly prepared ones.

The absence of an instant comes from the record criterion. The loss accumulates at the amplitude level cycle by cycle (§3.2), and accumulation is not an event; decay is completed when the loss turns into an escaping excitation carrying a record, and only when the record is established is there an event[3]. When the decay happened has no answer before the record, because before the record nothing happened. The macroscopic paradox triggered by a decay[32] dissolves accordingly: the record of the detector is the phase carried away irrevocably by a really emitted photon[3], and once the record is established everything downstream of the chain is at the probability level; what is in superposition is only the process at the amplitude level, never a macroscopic body that has been recorded.

The rate of decay is at the same time a clock of proper time. Proper time counts cycle phase, $\Delta\tau = \Delta\Phi/m$[3]; for a standing wave in motion the exchange share per cycle is suppressed by m/E[3], the transit sits on the formation-exchange step (§2.1), and the loss share is suppressed in the same proportion (to first order, $\varepsilon \ll 1$),

$$\varepsilon_{\text{motion}} = \frac{m}{E}\,\varepsilon\,, \qquad \tau = \frac{E}{m}\,\tau_0\,,$$

so the lifetime read in the laboratory is dilated by E/m. Cosmic-ray muons are produced more than ten kilometers up, the distance corresponding to the rest lifetime is less than seven hundred meters, and the count reaching sea level reads precisely this dilation[22].

The lifetime is the reciprocal of the rate; the instant of decay does not exist beforehand in the state, and the magnitude of the rate is the subject of §5.4.

5.2 The β Spectrum

The electron energy spectrum of β decay is continuous[20]. In a two-body decay the final energies are unique and the spectrum should be a single line; the measured spectrum extends instead from zero all the way to the endpoint, and the continuous spectrum once cast doubt on energy conservation itself. Pauli postulated an unobserved neutral particle for it[30], and Fermi built the theory of the contact interaction on that basis[8]. The emitted pair, the shape of the spectrum, and the endpoint are derived from the structure below.

The emitted pair comes from the restriction of the pairing by charge (§2.2): the emitted pair carries the integer charge difference between the two ends of the line, a charged lepton cannot constitute it alone, and its partner is necessarily a neutral lepton; the emitted pair of the β branch is $(e, \bar{\nu}_e)$, and the final state is always three-body; the emitted pair of a purely leptonic transit contains exactly one neutrino, so the energy spectrum of the muon is continuous as well. In the state space the final state of the transit is three standing waves; in observational spacetime what is read out is the energy distribution of the electron, the three-body division of energy has no unique solution, and the spectrum is therefore continuous. The same pairing rule gives a single line for a two-body final state: the final state of $\pi^\pm \to \mu\nu_\mu$ consists of the emitted pair alone, and the muon is monoenergetic (§4.7); the continuity of three-body and the single line of two-body spectra come from the same emitted pair. The neutrino here is what charge conservation requires of the emitted pair.

The shape of the spectrum comes from the rate formula. The strength of the transit is absorbed into the cycle rate (§4.3), the vertex is of contact form, the nuclear matrix element of an allowed transition carries no final-state momentum, the spinor contraction on the lepton side is given by the $V-A$ form (§4.2), and the phase space of the rate formula gives the whole shape of the spectrum[33]: the density of states on the electron side is $p_e E_e$, on the side of the massless neutrino $(E_0 - E_e)^2$, and the recoil balances momentum while taking almost no energy,

$$\frac{d\Gamma}{dE_e} \;\propto\; p_e\, E_e \left(E_0 - E_e\right)^2 , \qquad E_0 = \Delta m \, .$$

The endpoint is given by the mass difference: with $\Delta m = 1.30\,\mathrm{MeV}$ computed in reference [1] and $m_e = 0.511\,\mathrm{MeV}$, the endpoint kinetic energy is $0.79\,\mathrm{MeV}$, the experimental value being $0.782\,\mathrm{MeV}$[45]. The spectrum goes to zero quadratically at the endpoint, and the mass of the neutrino distorts the shape of the approach to zero,

$$\frac{d\Gamma}{dE_e} \;\propto\; p_e\, E_e\, (E_0 - E_e)\sqrt{(E_0 - E_e)^2 - m_\nu^2}\, ;$$

the neutrino mass spectrum computed in reference [1] thereby enters the shape of the endpoint, each mass eigenstate entering separately with its own threshold and its electron-flavor weight, the target of the endpoint measurements[29].

The particle historically inferred from the continuous spectrum is given directly by the charge structure of the emitted pair in the mechanism; its single winding direction appears in §5.3.

5.3 Parity Violation

Two newly discovered mesons had the same mass and the same lifetime yet decayed to final states of opposite parity; Lee and Yang examined the literature and pointed out that parity conservation in weak processes had never been tested by experiment[10]. The next year Wu and collaborators polarized ^{60}Co at a hundredth of a kelvin and found more electrons emitted opposite to the nuclear spin[11], and the asymmetry of the muon in the $\pi \to \mu \to e$ chain was observed at the same time[37]; reflected in a mirror, the apparatus should emit more electrons along the spin, and this mirror-image experiment does not occur in nature. The helicity of the neutrino (the reading of the conserved angular-momentum component along the direction of propagation[3]) was subsequently measured to be a single left-handed one[26]. This set of facts is derived one by one from the structure below.

The form of the mirror image in the readout is given by the readout geometry of reference [3]. The two sites of a doublet span $\mathbb{C}^2$, and the four-dimensional reading is its Hermitian form; the lattice reflection ι exchanges the two sites and acts on the reading as a rotation by π about an axis, a symmetry of the readout and not the mirror; spatial inversion is given by an antiunitary composite, the complex conjugation J followed by a rotation. The rotation is a symmetry, so the asymmetry under the mirror is the asymmetry under J. J maps a character to its conjugate, $J\chi_k = \bar{\chi}_k$, and the two members of a conjugate pair are the two winding directions of one standing wave[1]: the mirror reverses the winding direction.

The place of the asymmetry. The coefficients of the operators of propagation and transit are all real, they commute with J, and the modulus of an amplitude is unchanged under the mirror; the asymmetry lies in the states. Complex directions come in pairs, J maps one winding direction to the other, and a state involving a pair of complex directions has a mirror state; the real direction χ_3 has a single winding direction, J fixes it, and the mirror state of its winding direction does not exist,

$$J\chi_3 = \chi_3 \, , \qquad J\chi_k = \chi_{-k} \quad (k \neq 0, 3)\, .$$

The breaking direction χ_3 carries the W, and the direction of the neutrino is also χ_3 (§2.1): decays through a transit and final states containing a neutrino have no mirror counterpart, the neutral excitation of the weak channel $Z \to \nu\bar{\nu}$ among them[1]; strong and electromagnetic releases do not involve χ_3 and are symmetric under the mirror. The neutrino has a single winding direction and enters the emitted pair with a single helicity[26]. The contact matrix element takes the $V-A$ form[12, 38], the combination paired with the single-winding neutrino, which is the formal condition of this paper (§4.2).

The coefficient of the asymmetry is fixed by the weights. The β asymmetry of polarized neutrons is given by the ratio of the vector and axial-vector weights, and the angular distribution of the contact matrix element (§4.2) gives the coefficient of electron emission along the nuclear spin as

$$A_n = -\frac{2\, g_A(g_A-1)}{1+3g_A^2}\,;$$

with $g_A = 1.2746$ of §4.6 the tree-level coefficient is -0.1192; the leading coefficient extracted by experiment is -0.1199[43], corresponding to the $g_A = 1.2764$ of that experiment, and the difference is the residual of g_A amplified by the sensitivity of A_n to g_A.

The change of identity and the breaking of the mirror come from the breaking direction χ_3.

5.4 The Lifetime Hierarchy

The lifetimes of unstable states fall into tiers. Strong resonances release on the scale of 10^{-23} seconds, electromagnetic de-excitations on the scale of 10^{-17} seconds, and weak decays extend upward from 10^{-13} seconds; both being π mesons, the π^0 has a lifetime of 8.5×10^{-17} seconds and the $\pi^{\pm}$ one of 2.6×10^{-8} seconds, nine orders of magnitude apart[45]. The spacing of the tiers is derived from the structure below.

The first divide is the presence or absence of a transit. Release and de-excitation do not change identity, pass through no zero element, and have rates of first order in the channel coupling α_k (§6); a change of identity must pass through the second-order zero-element transit (§1.1), the contact coupling of the transit $G_F = 1/(\sqrt{2}\,v^2)$ (§4.3) enters the amplitude once, and the rate takes the modulus squared:

$$\Gamma_{\text{release}} \;\sim\; \alpha_k\, m\,, \qquad \Gamma_{\text{decay}} \;\sim\; m\left(\frac{m}{v}\right)^4.$$

For the muon, $(m_\mu/v)^4 = 3.4\times10^{-14}$: the transit lengthens the reference scale $1/m_\mu \sim 10^{-23}$ seconds by nearly fourteen orders of magnitude, the numerical factor of the phase space $384\pi^3 \approx 1.2\times10^4$ lengthens it by four more, and together they make up the 2.2×10^{-6} seconds of §4.3. The decay of π^0 emits no neutrino and is a release (§6); $\pi^{\pm}$ passes through a transit and is suppressed once more by the winding seed (§4.7): the nine orders of magnitude are the direct display of the same divide on the same meson.

The second divide is the fifth power of the available energy. Within one tier, the rates are distributed with the fifth power of the available energy, the dimension of $\Gamma \propto m^5/v^4$ being given by the rate formula (§4.3); Sargent first read the fifth power off the endpoint energies of β emitters[21]. Between μ and τ, $(m_\mu/m_\tau)^5 = 7.4\times10^{-7}$, the observed lifetime ratio is 1.3×10^{-7}, and the remaining share is the branch counting of the τ (§4.5).

The third divide is the shift of generation charge. Strange particles are produced in pairs in strong collisions yet have lifetimes of 10^{-10} to 10^{-8} seconds, thirteen orders of magnitude slower than the process producing them[23, 24]; Gell-Mann and Nishijima introduced a new quantum number, conserved by the strong interaction and changed by one unit in each weak decay[25, 35], and Cabibbo found that the amplitudes changing strangeness share one strength, differing only by a universal angle[13]. Strangeness in the mechanism is the generation charge (§2.1): the strong and electromagnetic vertices carry no lattice operator and cannot rewrite the generation record, so a second-generation quark can be produced only in a pair with its antiparticle, which is associated production; a change of identity has the transit as its only route, and a cross-generation change is read out further by the lattice projection of the transit as the share of the locked link,

$$|V_{us}| = \lambda = 0.2247\,, \qquad \lambda^2 = 0.0505 \approx \tfrac{1}{20}\,,$$

a transit through the link between the first and second generations suppressing the rate by a further factor of twenty; λ is the value computed in reference [1], and the share $\sqrt{1-\lambda^2}$ remaining in the same generation has been compared in §4.6; the link shares of the third generation ($|V_{cb}|$, $|V_{ub}|$) are not given by λ and are not among the inputs of this paper (§7.2). Cabibbo's universal angle is here the universality of the lattice readout: the share belongs to the structure of the transit, whatever line passes through it.

The fourth divide is two transits. A four-vertex process of two transits in series carries a further fourth power of a transit in the rate (the nuclear matrix element and the gap of the intermediate state being counted separately); double β decay thereby becomes the slowest decay ever observed directly, with half-lives of the order of 10^{20} years[27]. The two lines of K^0 and $\bar{K}^0$ each cross one generation, their mixing proceeds by the four-vertex process of two transits (§2.2), and the smallness of the mixing follows, historically named the $\Delta S = 2$ rule[14].

The top of the hierarchy is a rate identically zero. The muon has a single decay branch, and the electromagnetically easier $\mu \to e\gamma$ has never appeared: if an electromagnetic vertex could rewrite the generation, the share of the radiative branch would be counted by the coupling α, of order 10^{-2}; experiment has pushed the upper bound down by ten orders of magnitude to 4.2×10^{-13} without a single event[16], and $\mu \to 3e$ is equally absent[17]. The electromagnetic vertex carries no lattice operator and cannot rewrite the generation record, so the amplitude of $\mu \to e\gamma$ is identically zero; the lepton generations have no links, a lepton at the transit can only continue in its own generation, and the amplitude of $\mu \to 3e$ is likewise zero (§2.1); the generation charge leaves with ν_μ, e and $\bar{\nu}_e$ appear as the generation-neutral emitted pair, and the path is unique (§2.2). A neutral cross-generation process has no single-transit path, historically named the absence of flavor-changing neutral currents[14]. On the quark side the same conservation law crosses generations with the share λ; on the lepton side there is no link to read and the cross-generation reading is zero; the asymmetry of the two sides comes from the two constructions of the inter-generation structure[1]. A rate identically zero has no degree, and a single event suffices to refute it (§7.2); the stability of the proton is the same kind of zero (§5.5).

The spacing of the tiers is the count of transits and the readout of links, combined into one schematic power law,

$$\Gamma \;\propto\; m \left(\frac{m}{v}\right)^{4n} \lambda^{2k}\,,$$

n the number of transits, $n = 0$ being release, k the number of passages through the link between the first and second generations, and m the available energy of the process; the numerical factors of each process (phase space, the multiplication of branch probabilities in a cascade, the gap of a virtual intermediate state, the propagator of an on-shell W) are computed process by process. The tiers from 10^{-23} seconds to 10^{20} years are sorted thereby. The zero at the top is not on the tiers; it is an identity.

5.5 Nucleon Stability

The neutron decays with a lifetime measured in minutes; the proton is stable, with a lifetime bound of 2.4×10^{34} years[18]. Their masses differ by 1.4 parts in a thousand, yet the answers are opposite. The two answers and the magnitude of the neutron lifetime are derived from the structure below.

On the neutron side, the internal projection has exactly one open branch, and the mass difference $\Delta m = 1.30\,\mathrm{MeV} > m_e$ computed in reference [1] lets the final state be completed (§2.3). The magnitude of minutes comes from the fifth-power scaling (§5.4). With the muon as reference,

$$\frac{\Gamma_n}{\Gamma_\mu} = 96\, f_0\, (1 + 3g_A^2)\, |V_{ud}|^2 \left(\frac{m_e}{m_\mu}\right)^5 ,$$

where f_0 is the phase-space integral in units of m_e, $1 + 3g_A^2$ is the combination of the vector and axial-vector readouts (§4.2), and g_A and $|V_{ud}|$ take the values of §4.6: $(m_e/m_\mu)^5 = 2.6 \times 10^{-12}$ lengthens the lifetime by twelve orders of magnitude, $96\, f_0(1 + 3g_A^2)$ shortens it again by about three, and together they give the order of 10^3 seconds, the observed order of minutes[45]. The electron forms at, and leaves, the charged emergent state; the electromagnetic correction of the electron line is not derived in this paper, and the precise value of the lifetime is not among the outputs of this paper.

On the proton side, the branches in which a quark becomes a quark have heavier final states and are closed by the completion condition; the branches in which a quark becomes a lepton or a neutrino are closed by charge (§2.3), the charge difference being fractional while channel excitations carry only integer charge, so no emitted pair can balance it. The latter is the zero of an identity, with no degree. In the state space all branches of the proton are closed; in observational spacetime there is no event. The stability of every atomic nucleus rests on this zero and on the completion condition.

Two adjacent states on the spectrum, one with its branch just open, the other with all branches closed; the divide is only 1.4 parts in a thousand.

The five phenomena come from one structure. The exponential decay law gives the boundary of records, the β spectrum gives the emitted pair, parity violation gives the single winding direction of the real direction, the lifetime hierarchy gives the powers of the transit and the shift of generation charge, and nucleon stability gives the two faces of completion and closure. Every quantitative reading comes from the evaluation at the same vertex; the definition of decay has withstood the tests from the release at 10^{-23} seconds to the double β decay at 10^{20} years.

6 The Classification of Decays

In standard usage every spontaneous passage of an unstable state to a lower state is called a decay, and decays are divided by interaction into strong, electromagnetic, and weak. The preceding sections have established the class that changes identity, corresponding to weak decay. The three classes are distinguished within one cycle structure: each class has its own source of loss, and the three share the loss share per cycle and the readout structure of §3.

6.1 Three Classes of Decay

The amplitude of an unstable state is lost cycle by cycle. Of the three steps of the cycle, the formation-exchange step trades amplitude with the background and the pattern-reconstruction step fixes the position (§1.1); the exchanged amplitude either merges into the background through the zero element or falls back along a channel, and the reconstruction either stays in place or moves beyond a barrier. Through the zero element, back along a channel, or beyond a barrier: these are the only three sources of spontaneous loss.

The first class is zero-element transit. The amplitude of the initial state merges into the background through the zero element, the final state re-forms from the background, identity changes, and the emitted pair carries the record away (§2); the transit is a second-order process, and its rate carries the suppression $(m/v)^4$ (§4). The charged-current weak decays belong to this class; β decay and the decays of μ, τ, K mesons, $\pi^{\pm}$, and W are its instances.

The second class is resonance release. A channel excitation returns its weight to the channel, the identity of the standing wave is unchanged, and no zero element is passed; the strength of the fall-back is of first order in the channel coupling α_k, given by the resonance structure of the vertex direction k[1]. Strong and electromagnetic decays belong to this class; $\rho \to \pi\pi$, $\pi^0 \to \gamma\gamma$, the γ de-excitation of nuclei, and the decay of Z are its instances.

The third class is **barrier rearrangement**. Constituents cross a barrier as a whole along the multiplicative direction and are rebuilt on the far side; the identity of each constituent is unchanged, and no zero element is passed; the crossing probability is suppressed by $e^{-\beta d}$ with d the metric width of the barrier[3]. α decay and spontaneous fission belong to this class.

Source of loss	Zero element	Identity	Suppression factor of the rate	Instances
zero-element transit	yes	changed	$(m/v)^4$	β, μ, τ, K, $\pi^{\pm}$, W
resonance release	no	unchanged	α_k	$\rho \to \pi\pi$, $\pi^0 \to \gamma\gamma$, γ de-excitation, Z
barrier rearrangement	no	unchanged	$e^{-\beta d}$	α decay, spontaneous fission

The three classes are distinguished structurally: whether the zero element is passed, whether identity changes, and whether the position moves as a whole.

The three classes share one readout structure. The loss share per cycle ε enters the readings of §3.3; the exponential decay law, the Breit–Wigner line shape, and the branching ratios hold alike for the three classes, and the difference lies only in the source of ε and in the suppression factor:

$$\frac{\Gamma_{\text{transit}}}{m} \;\propto\; \left(\frac{m}{v}\right)^4 \times (\text{phase-space factor}), \qquad \frac{\Gamma_{\text{release}}}{m} \;\propto\; \alpha_k\,, \qquad \varepsilon_{\text{rearrangement}} \;\propto\; e^{-\beta d}\,.$$

The fast end of the lifetime hierarchy (§5.4) is resonance release and the slow end is zero-element transit; barrier rearrangement spans the range between them, its exponential dependence making its lifetime vary by orders of magnitude with the energy difference. The rates of the latter two classes are given in the following two subsections.

6.2 Resonance Release

Resonance release is the fall-back of a channel excitation. Resonances and excited states are channel excitations on top of a stable standing wave; the convolution operator K is diagonal in the character basis, and the weight of the excitation flows back along the same channel, passing through no zero element and changing the identity of no standing wave (§2.2). The fall-back proceeds once per cycle, and the final state is an excitation of the channel itself: the strong channel gives hadrons, and the electromagnetic channel gives photons.

The rate of release is given by a first-order vertex. The coupling of the channel is $\alpha_k = |R_k|^2/N_k^{\text{eff}}$[1], and the matrix element of the fall-back contains a single vertex,

$$|M|^2 = c\,\alpha_k\,m^2\,,$$

with m the mass of the excitation and c the structure factor of angular momentum and isospin; inserted into the golden rule in continuous time (§4.2) with the two-body phase space $\rho_2 = p_f/(8\pi m^2)$, p_f the final-state momentum,

$$\Gamma_{\text{release}} = |M|^2\rho_2 = \frac{c\,\alpha_k}{8\pi}\,p_f \;\sim\; \alpha_k\,m\,, \qquad \varepsilon_{\text{release}} = \frac{\Gamma_{\text{release}}}{v} \;\sim\; \alpha_k\,\frac{m}{v}\,.$$

The matrix element of a zero-element transit contains the contact coupling $G_F = 1/(\sqrt{2}v^2)$, and the rate carries $(m/v)^4$ on squaring (§5.4); the matrix element of a release contains no contact coupling, the generation charge does not shift, and the combined formula of the lifetime hierarchy gives the same result at $n = 0$. The widths of strong resonances are thereby measured in hundreds of MeV; the ratio of width to mass of $\rho \to \pi\pi$ is 0.19, of the same order as $\alpha_s^{(0)} = 0.115$; the electromagnetic coupling $\alpha_{\rm EM}^{(0)} = 0.0079$ is smaller by more than an order of magnitude, the widths of γ de-excitations are suppressed accordingly, and $\pi^0 \to \gamma\gamma$ contains two electromagnetic vertices and is of second order in $\alpha_{\rm EM}$.

The release of the Z is an instance of the same type. Z belongs to an even direction and is a channel excitation (§2.2); its width adds term by term over the open directions,

$$\Gamma_Z = \sum_f \Gamma(Z \to f\bar{f})\,, \qquad \Gamma_{\rm invisible} = N_\nu\,\Gamma(Z \to \nu\bar{\nu})\,,$$

and the invisible share counts the number of neutrino generations $N_\nu = (p-1)/2 = 3$[1], the LEP count being $N_\nu = 2.984 \pm 0.008$[28]. The width of a release counts directions and carries no transit suppression; the rate of barrier rearrangement comes from yet another source.

6.3 Barrier Rearrangement

Barrier rearrangement is a displacement of position as a whole. A part of a composite body, an α cluster or a fission fragment, moves as a whole from inside a barrier to outside it; the standing wave and identity of each constituent are unchanged, and only the position changes. Reconstruction inside the barrier region is suppressed, and site-by-site motion along the additive direction cannot pass; one multiplicative step does not pass site by site but connects directly two positions differing by a factor p, the suppression does not act on it, and the pattern is rebuilt directly on the far side of the barrier[3].

The rate of rearrangement is given by the metric width of the barrier. In the metric $d(x,y) = |\log(x/y)|$ the KMS weight $n^{-\beta} = e^{-\beta\,d(1,n)}$ is the exponential of the metric distance[1]; one multiplicative step $x \mapsto px$ crosses the metric distance $\log p$ with weight $p^{-\beta}$; the amplitude share passing through the multiplicative channel per cycle is suppressed by the metric width d of the barrier region, and the probability that the event lands on the far side is the modulus squared by the Born rule[3],

$$A \;\propto\; e^{-\beta d/2}\,, \qquad \varepsilon_{\rm rearrangement} = C\,e^{-\beta d}\,, \qquad \Gamma = v\,C\,e^{-\beta d}\,, \qquad \log\tau = \beta d - \log(vC)\,,$$

with C the reconstruction factor in front of the barrier. The metric width of the barrier grows as the available energy E decreases; for a Coulomb barrier (the potential given in reference [4]), with the metric width computed by the exponential law of equal width with WKB[3], the exponent of the crossing varies with $Z/\sqrt{E}$, in agreement with Gamow's result[6], so the logarithm of the lifetime is linear in $Z/\sqrt{E}$, the Geiger–Nuttall law[7]: the α half-lives of a family of nuclides extend from below a microsecond to 10^{19} years, more than thirty orders of magnitude corresponding to an energy difference of a few MeV. This span comes from the exponential dependence and belongs to a source different from the transit powers of §5.4; spontaneous fission is likewise a rearrangement of position.

The derivation of the crossing is established in the mechanism of motion[3], and this paper takes its result. The loss shares of the three classes enter one readout; the numbers and predictions of this paper are collected in the next section.

7 Conclusion

The numerical outputs and structural corollaries of this paper are compared with experiment item by item, and the amplitudes identically zero give falsifiable predictions.

7.1 Numerical Summary

The numerical outputs compared with experiment are as follows; all inputs on the side of this paper are computed outputs of reference [1], and the experimental column is taken from reference [45] and the individual experiments.

Quantity	This paper	Experiment	Deviation
G_F [GeV^{-2}]	1.16665×10^{-5}	1.16638×10^{-5}	+0.023%
τ_μ [s]	2.18709×10^{-6}	2.19698×10^{-6}	−0.45%
$\Gamma(\tau \to e)$ [GeV]	4.050×10^{-13}	4.040×10^{-13}	+0.25%
$\Gamma(\tau \to \mu)$ [GeV]	3.939×10^{-13}	3.943×10^{-13}	−0.10%
$\mathrm{BR}(\tau \to \mu)$	0.1732	0.1739	−0.40%
τ_τ [s]	2.894×10^{-13}	2.903×10^{-13}	−0.30%
$\lvert V_{ud}\rvert$	0.97443	0.97367	+0.078%
g_A	1.2746	1.2754	−0.06%

The table contains no inherited correction. The electromagnetic correction of the charged lepton lines is not derived in this paper; τ_μ and the leptonic branches of the τ do not contain this share, which in the standard calculation is -0.42% (§4.3). Beyond the numbers, there are nine points of agreement at the structural level:

- the exponential decay law is a corollary of the cycle readout (§3.3);
- the Breit–Wigner line shape is the resolvent of the sum over cycles (§3.3);
- the $V{-}A$ pairing is the contact form paired with the single-winding neutrino, the formal condition of this paper (§5.3)[12];
- the strength of the transit is universal over all directions, a corollary of the recorded position (§2.1);
- the prohibition of flavor-changing neutral currents is given by the readout structure of generation (no links between lepton generations, §2.1);
- the conservation of the vector current and the constancy of Ft are given by the commutator of μ_2 (§4.6);
- that the axial-vector weight of leptons is one is given by the absence of the contact configuration (§4.6);
- the visible and invisible classes of the Z width are given by the evenness of the channel excitation (§2.2);
- the direct coupling of the neutrino to the photon bath is excluded by electric neutrality (§2.3).

7.2 Testable Predictions

The following amplitudes are identically zero within the vertex catalogue of this paper (one line and one pair, and their compositions), and the observation of any one of them refutes it:

- $\mu \to e\gamma$ (conservation of generation charge);
- $\mu \to 3e$ (no links in the inter-generation structure of leptons);
- proton decay (closure by charge);
- n–$\bar{n}$ oscillation (the vertices conserve the net number of lines; $\Delta B = 2$ has no carrier)[19].

Two derivations remain to be completed. First, the electromagnetic correction of the charged lepton lines: once it is derived, the -0.45% of τ_μ should disappear. Second, the hadronic branches of the W: once the equal-share counting law is established, $\mathrm{BR}(W \to \ell\nu)$ becomes a numerical output of the framework, and the correction should move the structural value 1/9 toward the observed 0.1086.

Two phase structures lie outside the scope of this paper. CP violation in the cross-generation phases (the K and B meson systems[39]) and the strong CP problem belong to the part of the generation-structure phases not yet completed; the phase inputs of this paper stop at δ_2 and λ of reference [1], two real inputs cannot carry a CP phase angle nor give the link shares of the third generation, and the boundary of this paper is drawn here. The Majorana nature of the neutrino[34] and the criterion for neutrinoless double β decay are among the predictions of reference [1] and are not repeated here.

7.3 Concluding Remarks

Decay is a change of identity and proceeds through zero-element transit. The first-order matrix element for changing identity is identically zero and the second-order one is in general nonzero, so identity cannot be rewritten directly and can only merge into the zero element first and re-form from the background afterward. The transit is equivalent for all directions, and all selection comes from the evaluation at the two vertices: the two conservation laws of electric charge and generation charge restrict the change of identity to the two kinds $d \leftrightarrow u$ and $\ell \leftrightarrow \nu_\ell$, and the states with all branches closed are stable matter.

A process becomes an event in observational spacetime only through a record. The record of a decay is the difference of identity, carried by the emitted standing waves, and the transit is complete when the final state is completed and the carrier has escaped. The position and width of the spectral line and the count ratio of the branches come from one place: the former two are the phase and modulus of the cycle eigenvalue, the latter the distribution of the loss share.

Decay runs in one direction only, because merging into the zero element has no reverse, the occupation after the loss does not rise again, and the escape of the carrier cannot be undone. The rate is converted from the loss share per cycle, and the lifetime hierarchy is sorted into tiers by the powers of the transit; the Fermi constant, the lifetimes of μ and τ, the branching ratios, $|V_{ud}|$, and the axial-vector weight are given thereby, with all inputs quantities fixed beforehand in reference [1].

The decay defined in this paper is the change of identity, corresponding to weak decay in the standard classification; strong decay, electromagnetic decay, and α decay are, in this framework, resonance release and barrier rearrangement, which do not change identity; the three classes share one readout structure and differ in the source of loss. With this, the mechanisms of the formation, motion, thermalization, and decay of matter have each been established on the same state space[1, 3, 2]. The four processes share one cycle: formation builds the standing wave in the cycle, motion moves the reconstruction center with the cycle, thermalization relaxes the occupations of many standing waves through the zero element, and decay replaces the identity of a single standing wave through the zero element. They are four ways in which one cyclic exchange proceeds, and their unified dynamics is the work to come.

Data Availability

No data sets were generated or analyzed in this study; all results follow from analytic derivation.

Conflict of Interest

The author declares no conflict of interest.

A Verification of the Reflection Structure and the Record Resolution

The operator content and the process-by-process check of the reflection structure (§2.1) are as follows.

The parity reflection P is defined by $(Pf)(x) = f(-x)$. Characters are eigenfunctions of P, and multiplication operators and rank-one operators are accordingly eigenoperators of P:

$$P\chi = \chi(-1)\,\chi\,, \qquad P\,M_\chi\,P = \chi(-1)\,M_\chi\,, \qquad P\,|\chi\rangle\langle\chi'|\,P = \chi(-1)\chi'(-1)\,|\chi\rangle\langle\chi'|\,,$$

the zero element is fixed ($P\delta_0 = \delta_0$), K commutes with P, and the transit matrix element satisfies

$$P\,[D_F, T_a]\,P = [D_F, T_{-a}]\,.$$

The reflection does not exchange the two members of a conjugate pair, $P\chi_1 = -\chi_1$, $P\chi_5 = -\chi_5$; what exchanges them is the complex conjugation J, $J\chi_1 = \chi_5$; the record $|\chi_1\rangle\langle\chi_1| - |\chi_5\rangle\langle\chi_5|$ distinguishing the two species is P-even.

The two sectors of the record resolution are checked with the normalized directions $|k\rangle = \chi_k/\sqrt{p-1}$. The coherent absorption $\sum_a\langle\delta_a|k\rangle = 0$ for all $k \neq 0$; a record resolving the lattice site gives $\sum_a |\langle\delta_a|k\rangle|^2 = 1$ for all k; a record resolving only the doublet reads out with $|D_j\rangle = (\delta_{a_j} + \delta_{-a_j})/\sqrt{2}$, and $\sum_j |\langle D_j|k\rangle|^2$ is 1 for even k (1/3 per doublet) and 0 for odd k. Equivalently, with symmetric position weights $a, b \in \{\pm 1\}$ of equal weight, the second-order amplitude $\sum_{a,b}\langle\chi_k|[D_F, T_b][D_F, T_a]|\chi_j\rangle$ is nonzero if and only if χ_j and χ_k are both even.

Taking the transit of the muon as an example, the two conservation laws are checked item by item:

$$\mu^- \;\to\; \nu_\mu + (e^-, \bar\nu_e) \;: \qquad \underbrace{j(\nu_\mu) = j(\mu)}_{\text{generation charge}}\,, \qquad \underbrace{-1 = 0 + (-1+0)}_{\text{charge}}\,,$$

the line continues in its generation, the emitted pair is generation-neutral, and the path is unique.

For each transit the product of the direction parities, whether an odd direction is contained (if so, the record must resolve the site within the doublet), and the verdict are listed:

Process	Parity product	Odd direction	Verdict
$\mu \to \nu_\mu\,(e, \bar{\nu}_e)$	$+$	yes	allowed; exists
$d \to u\,(e, \bar{\nu}_e)$	$-$	yes	allowed; exists
$\tau \to \nu_\tau\,(q, \bar{q}\,')$	$-$	yes	allowed; exists
$s \to u\,(d, \bar{u})$	$+$	yes	allowed; exists
$s \to u\,(e, \bar{\nu}_e)$	$-$	yes	allowed; exists
$\mu \to \nu_\mu\,(e^+, e^-)$	$-$	yes	forbidden by charge ($-1 \neq 0$); absent
$\mu \to \nu_\mu\,(\nu, \bar{\nu})$	$-$	yes	forbidden by charge; absent
$\mu \to e\,(e^+, e^-)$	$+$	no	allowed by charge; forbidden by generation charge (§2.1)
emitted pairs of type $(q, \bar{\ell})$	—	—	forbidden by charge conservation

B Configurations of the Dwell Correction

The configuration count of the dwell correction (§4.4). The insertion pair is $\{\chi_2, \chi_4\}$ (mutually conjugate single-factor generators); both lines of the emitted pair are quark species. The configurations of each composite are the combinations (member, line), four in all; the intermediate residence is the position of the line multiplied by the conjugate of the emitted member. Taking the $(d, \bar{u})$ pair as an example ($\bar{u}$ being the antiparticle of χ_1, its direction factor the conjugate χ_5, so that both lines are read out in the χ_5 direction):

Member	Line	Intermediate residence	Contribution
χ_4	line 1	χ_1 (colored)	nonzero
χ_2	line 1	χ_3 (color-neutral, electrically neutral)	zero
χ_4	line 2	χ_1 (colored)	nonzero
χ_2	line 2	χ_3 (color-neutral, electrically neutral)	zero

The normalization is shared equally over the unitary displacement algebra, each of the four configurations taking one quarter; the vertex coupling of a neutral intermediate residence is zero (the generalization of the electric-neutrality exemption), and the nonzero configurations total one half, the share of Step 3 of §4.4. When the same count is applied to the Cabibbo correction of reference [1], the intermediate residences of both configurations carry charge (χ_1 and χ_5), the nonzero configurations total one, in agreement with the coefficient there; applied to the electrically neutral neutrino, all configurations are zero, returning to the electric-neutrality exemption itself. The three cases come from one rule. Every layer of the nesting (Step 5 of §4.4) repeats the same count: the object of the recursion is the segment, and the configuration space of a segment does not change with the layer.

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